VARSHA

AI-Powered Parametric Insurance for Delivery Partners

PRESENTED BY

TEAM SYNC

1. Executive Summary

VARSHA is an AI-enabled parametric insurance platform designed for food delivery partners on Swiggy and Zomato in Chennai. The platform protects gig workers from income loss caused by uncontrollable external disruptions including heavy rainfall, extreme heat, urban flooding, and curfews.

Core innovations

#	Innovation	Description
1	**Pay-as-you-earn premium model**	₹3 deducted per delivery with a weekly cap of ₹75, ensuring complete affordability.
2	**Shield accumulation mechanism**	Partners pay until ₹300, then receive free coverage until a payout occurs.
3	**5-signal fraud detection**	Multi-layered verification combining GPS, cell tower, IP, motion data, and IMD weather.

Key differentiator: Unlike traditional insurance that pays after 7–30 days, VARSHA delivers ₹2,000 within 90 seconds of trigger activation with zero paperwork.

2. Problem Statement

2.1 The Gig Worker Challenge

Chennai has over 50,000 active food delivery partners across Swiggy and Zomato. These workers face multiple disruptions that directly impact their income:

Disruption	Impact on earnings	Frequency in Chennai
Northeast Monsoon rains	2–3 days lost per event	Oct–Dec · 15–20 events / year
Summer heat (>40 °C)	Work stoppage during peak hours	April–June · 30+ days / year
Urban flooding	Zone closures	5–10 events / year (Velachery, Madipakkam)
Curfews / strikes	Complete work halt	2–5 events / year

2.2 Market Gap Analysis

Factor	Traditional insurance	What partners need
Premium	₹400–500 upfront weekly	Pay only when working
Payout time	7–30 days	Instant — same day
Claim process	Paperwork, branch visits	Zero-touch, automatic
Trust	"They will reject my claim"	Transparent, verifiable
Coverage	Single risk (accident / health)	Income protection only

2.3 The Karthik Persona

Karthik, 32 — Swiggy delivery partner, Velachery. Three years on the platform. Wife and daughter at home. Earns ₹600–800 daily across 25–30 orders per week. Daughter's school fees are due at month start. When rain floods Velachery he loses 2–3 days of work. He cannot afford ₹400 upfront. He cannot wait 7 days for a payout. He has no time for paperwork.

VARSHA is built for Karthik.

3. Solution Overview

3.1 Core Concept

VARSHA is a parametric insurance platform where payouts are triggered automatically when predefined conditions are met, eliminating claims processing entirely.

The VARSHA Promise

Pay only when you work	**Pay until covered, then free**
Get paid instantly when disruption hits	**Never worry about renewal**

3.2 Complete Workflow

Step	Stage	Details
1	**Onboarding (2 min)**	Mobile number registration · Zone selection (Velachery, Adyar, OMR, etc.) · UPI ID linking · No documents required
2	**Daily deductions**	₹3 deducted per successful delivery · Weekly cap of ₹75 · Dashboard shows shield progress · Partner barely notices the deduction
3	**Shield accumulation**	Shield target: ₹300 (~4 weeks / 100 deliveries) · After ₹300 deductions STOP · Partner fully covered at zero cost
4	**Parametric trigger**	IMD detects >20mm rain in 6 hrs · OR temperature >40 °C · OR flood sensors trigger · OR curfew announced
5	**Fraud detection**	GPS location vs zone claim · Cell tower matching · IP geolocation · Motion data (accelerometer) · IMD weather station — all five must align
6	**Auto-payout**	₹2,000 credited to UPI wallet in 90 seconds · Policy terminates · Zero forms, calls, or waiting · Transaction logged
7	**Auto-renewal**	Next delivery triggers new ₹3 deductions · Coverage never lapses · Partner never needs to remember
8	**RainBoost**	First 3 orders after payout pay DOUBLE · Encourages return to work · Helps catch up lost income · Fraud detection prevents collusion
9	**TipPoints**	Customer tips during hardship earns loyalty points · Partner gets full tip instantly · Cash tips verified via 50m proximity check

4. Premium Model

4.1 Pay-as-You-Earn Structure

Component	Value	Rationale
Base deduction	**₹3 per order**	Barely noticeable; aligns directly with earnings
Weekly cap	**₹75**	Never exceeds 25 orders / week of payment
Shield target	**₹300**	~4 weeks of work for full-time partners
Payout amount	**₹2,000**	Covers 3–4 days of lost wages
Risk multiplier	**0.8× – 1.5×**	Fair pricing based on zone flood-risk profile

4.2 Zone-Based Risk Factors — Chennai

Zone	Risk level	Multiplier	Per-order deduction	Weekly cap
Velachery	High — frequent flooding	**1.5×**	**₹4.50**	₹75
Madipakkam	High	**1.4×**	**₹4.20**	₹75
Adyar	Medium	**1.2×**	**₹3.60**	₹75
T.Nagar	Medium	**1.0×**	**₹3.00**	₹75
OMR	Low	**0.9×**	**₹2.70**	₹75
Anna Nagar	Low	**0.9×**	**₹2.70**	₹75

4.3 Shield Accumulation Example — Karthik (Velachery, 1.5× multiplier)

Week	Orders	Deduction	Cumulative	Status
1	25	₹75 (capped)	**₹75**	Building shield
2	28	₹75 (capped)	**₹150**	Building shield
3	30	₹75 (capped)	**₹225**	Building shield
4	25	₹75 (capped)	**₹300**	**SHIELD FULL**
5	Rain event	₹0	**₹300**	**₹2,000 payout**
6	After payout	₹75	**₹75**	New cycle begins

4.4 Fairness Analysis

Scenario	Traditional insurance	VARSHA
Works 30 orders / week	₹400 fixed	**₹75**
Works 15 orders / week	₹400 fixed	**₹45**
Works 0 orders (sick)	₹400 still due	**₹0**
After 4 weeks (shield full)	Still paying	**Free until payout**

5. Parametric Triggers

5.1 Trigger Definitions

Trigger	Threshold	Data source	Update frequency
Heavy rainfall	**>20mm in 6 hours**	IMD Chennai	Every 3 hours
Extreme heat	**>40 °C**	IMD Chennai	Every 6 hours
Urban flooding	**Water level >1.5m**	Chennai Corporation	Every 30 minutes
Curfew / strike	**Government notification**	Official API + manual	Real-time

5.2 Why 20mm in 6 Hours?

Analysis of Chennai weather data (2015–2025) reveals a clear threshold effect:

Rainfall level	Operational impact
< 15mm	Deliveries continue with a slight slowdown — no material income loss
15–20mm	Waterlogging begins · 30% delivery drop — borderline threshold
20–25mm	Major roads affected · 70% delivery drop — trigger activates
> 25mm	Complete halt in low-lying areas (Velachery, Madipakkam) — all income lost

Design rationale: 20mm is the threshold where partners genuinely cannot work, but the trigger activates frequently enough to provide meaningful protection throughout the monsoon season.

5.3 Data Integration Architecture

All triggers use government and commercial APIs in a redundant chain:

	Source	Role	Trigger covered
1	**IMD Chennai API**	Primary weather source	Rainfall + heat
2	**OpenWeatherMap API**	Backup / redundancy	Rainfall + heat
3	**Chennai Corporation flood sensors**	Hyper-local flood data	Urban flooding
4	**News API + Govt Alerts**	Social disruption alerts	Curfew + strikes

6. Fraud Detection System

6.1 The 5-Signal Coherence Model

VARSHA does not trust any single signal. Every claim is validated against five independent truth sources simultaneously:

Signal	Data source	Spoofable?	VARSHA mitigation
GPS	Phone location	**Yes**	Cross-check with cell tower and IP
Cell tower	Network API	**No**	Fixed infrastructure — cannot be faked
IP geoloc.	Network provider	**Partially**	Must match tower location exactly
Motion	Accelerometer	**Partially**	Pattern analysis vs. typical delivery
Weather	IMD station data	**No**	Independent of phone; zone-specific

6.2 Storm Coherence Score (SCS)

Each signal receives a match score (0–1). The five scores are combined using a weighted formula:

```
SCS = (0.25 × LocationMatch) + (0.20 × MotionMatch) + (0.20 × PlatformMatch)
            + (0.20 × WeatherMatch) + (0.15 × NetworkMatch)
```

Decision matrix based on the computed SCS:

SCS range	Decision	Timeframe	Experience for partner
0.80 – 1.00	**Auto-approve**	90 seconds	₹2,000 lands instantly. No action needed.
0.60 – 0.79	**Manual review**	Within 24 hours	Soft hold, partner notified, no penalty.
0.40 – 0.59	**Hold — can appeal**	24-hr window	Partner can provide additional context.
0.00 – 0.39	**Auto-reject + flag**	Immediate	Fraud pattern logged to ML model.

6.3 Real-World Fraud Scenarios

#	Scenario	Attempt	VARSHA detection	Outcome
1	**GPS spoofer (Rajesh, Noida)**	Spoof GPS to Velachery from Noida	Cell tower shows Noida · IP shows Mumbai	SCS 0.22 → Auto-reject + account flagged

2	**Syndicate ring (12 phones, café)**	12 accounts claim simultaneously · all in Velachery	RDI 0.71 · same WiFi · same referral · same UPI bank	Cluster quarantine · all claims held for review
3	**RainBoost colluder (Mani, tiffin shop)**	Cousin places 3 fake orders · Mani collects double pay	Order diversity 0 · restaurant baseline anomaly	RainBoost cancelled · restaurant trust score drops
4	**Tip loop fraud (Kumar)**	Friend confirms ₹50 cash tip remotely	Proximity check fails — friend's phone 3km away	Tip rejected · soft warning logged

6.4 Ring Density Index (RDI)

The RDI detects multi-phone fraud rings by analysing six weighted graph signals:

Signal	Weight	Detection method
Same WiFi BSSID	**20%**	Network fingerprinting across claims
Same onboarding IP	**15%**	IP clustering at registration
Same referral chain	**20%**	Social graph analysis
Claim timing within 4 minutes	**15%**	Temporal clustering
Same UPI payout bank	**20%**	Transaction destination tracing
Identical motion patterns	**10%**	Accelerometer signature comparison

RDI value	Action
≥ 0.70	Cluster quarantine — all claims in the cluster held pending human review
0.50 – 0.69	Flag for review — individual accounts monitored
< 0.50	Normal variation — no action

6.5 Tip Proximity Verification

Check	Requirement
Proximity	Customer phone within 50m of delivery GPS at confirmation time
Timing	Confirmation must occur within 24 hours of delivery completion
History	Partner's cash tip ratio benchmarked against zone average
Pattern	Multiple confirmations from same customer flagged for review

> **Rule:** If the proximity check fails, the tip is automatically rejected regardless of all other factors. Cash tips must happen face-to-face.

7. AI / ML Architecture

7.1 Model Overview

Model	Purpose	Algorithm	Training data
Risk scoring	Premium calculation	Random Forest / XGBoost	10 years IMD Chennai data
Fraud detection	Claim validation	Isolation Forest + XGBoost	50,000 synthetic patterns
Ring density	Syndicate detection	Geospatial clustering	WiFi + IP + timing data
Predictive disruption	Forecasting	LSTM (time series)	IMD forecasts + historical
Order diversity	RainBoost fraud	Regression + baseline	Restaurant 30-day history

7.2 Risk Scoring Model

Input features

- Historical rainfall by zone (10 years)
- Flood incidence records (Chennai Corporation)
- Partner order history (consistency score)
- Time of year (monsoon indicator)
- Zone elevation data (flood risk)

Output and accuracy

Output	Algorithm	Validation accuracy
Risk factor multiplier (0.8× – 1.5×)	XGBoost	**94.3% (5-fold cross-validation)**

7.3 Fraud Detection Model

Input features

- GPS vs. cell tower mismatch score
- Accelerometer variance (motion pattern)
- Claim timing density (per zone per hour)
- Swiggy order acceptance rate
- Device fingerprint consistency

Anomaly detection	Classification	Fraud flag threshold
Isolation Forest	XGBoost classifier	**95th percentile score**

7.4 Predictive Disruption Model (LSTM)

Input features

- IMD 7-day forecast (rainfall, temperature)
- Historical flood patterns by date
- Traffic density (MapMyIndia)
- Social media alerts (bandh detection)

Output	Architecture	Window
Disruption probability (0–1) per zone	**LSTM**	7-day sliding window

7.5 Continuous Learning

All models retrain on a weekly cycle using the following data streams:

Data stream	Contribution to model improvement
New claim data	Refines SCS weights; improves auto-approve accuracy
New fraud patterns	Updates Isolation Forest baseline; tightens RDI clusters
Updated weather data	Recalibrates LSTM sliding window; adjusts zone multipliers
Partner feedback	Validates false-positive rate; informs UX adjustments

8. Platform Architecture

8.1 Technology Stack

Layer	Technology	Rationale
Frontend	**React + Tailwind CSS**	Fast development, mobile-first, PWA capable
Backend	**FastAPI (Python)**	Async support, easy ML integration, high performance
Database	**PostgreSQL**	ACID compliance for financial transactions
Cache	**Redis**	Real-time trigger handling, session storage
Queue	**Celery**	Async payout processing, fraud detection jobs
ML	**scikit-learn + XGBoost**	Production-ready, interpretable, well-documented models
Weather	**IMD + OpenWeatherMap**	Primary + backup redundancy for trigger reliability
Maps	**MapMyIndia**	Chennai-specific mapping; flood zone overlays
Payments	**Razorpay (mock / sandbox)**	UPI integration; test mode for Phase 1

8.2 Data Flow

#	Event	System flow
1	**Partner registers**	Input → PostgreSQL (users) · UPI ID stored encrypted
2	**Each delivery**	Event → Redis (daily counter) → PostgreSQL (transactions)
3	**Weather API poll**	Celery task (trigger check) → Redis alert if threshold met
4	**Trigger condition met**	Fraud detection (ML pipeline) → PostgreSQL (claim record)
5	**Claim approved**	Razorpay payout API → Partner UPI wallet · 90-second SLA
6	**All events**	Streamed to analytics dashboard in real-time

9. Analytics Dashboard

9.1 Partner View — Key Widgets

Widget	Information displayed
Shield status	Progress bar showing ₹ accumulated vs ₹300 target · deliveries this week · amount paid
Next payout	Trigger condition (e.g. rain >20mm in 6 hrs in your zone) · current risk level
History	Total protected this year · date of last payout
Forecast	Tomorrow's weather probability · active coverage status
RainBoost	Remaining double-pay orders (of 3) after a payout event

9.2 Admin / Insurer View — Key Widgets

Metric	Example value	Purpose
Active policies	**1,247**	Live coverage count
Premium collected (MTD)	**₹1,24,700**	Revenue tracking
Payouts (MTD)	**₹84,000**	Claims cost monitoring
Loss ratio	**67.4%**	Commercial viability check
Today's triggers	**38 claims**	Zone-level breakdown (Velachery 12, Adyar 7…)
Fraud alerts	**2 active**	RDI cluster flags + ML model accuracy
ML model accuracy	**94.3%**	Live model performance
Predictive alert	**85% rain tomorrow**	Pre-emptive partner notifications

10. Development Plan

Phase	Period	Theme & activities	Key deliverables
Phase 1	Weeks 1–2 Mar 4–20	Ideate & Know Your Delivery Worker · Weather data collection · Flood zone mapping · Partner interviews (15+) · Architecture design · ML prototype planning	README · Zone risk map · Validated personas · Architecture diagrams · ML model specs · 2-min demo video
Phase 2	Weeks 3–4 Mar 21–Apr 4	Protect Your Worker · React PWA frontend · Registration + UPI linking · ₹3/order deduction logic · Weekly cap enforcement · Shield accumulation · Mock UPI payout	Functional PWA · Premium engine with cap · Working mock payouts · User registration flow
Phase 3	Weeks 5–6 Apr 5–17	Perfect for Your Worker · 5-signal fraud detection · ML model integration · Ring density engine · RainBoost · TipPoints · Partner + admin dashboards	Full fraud detection · Production-ready platform · 5-min demo video · Final pitch deck

11. Competitive Analysis

11.1 Feature Comparison Matrix

Feature	VARSHA	Bajaj ClimateSafe	ShieldRide	RideShield	SEWA	Digit
Pay-as-you-earn premium	Yes	No	No	No	No	No
Shield accumulation	Yes	No	No	No	No	No
Multi-trigger (4 types)	Yes	Partial	Partial	Partial	No	No
Instant payout (<2 min)	Yes	No	Yes	No	No	Yes
5-signal fraud detection	Yes	No	No	No	No	No
Ring Density Index	Yes	No	No	No	No	No
RainBoost	Yes	No	No	No	No	No
TipPoints	Yes	No	No	No	No	No
Proximity-verified cash tips	Yes	No	No	No	No	No
Auto-renewal	Yes	No	No	No	No	No
Chennai zone optimisation	Yes	No	No	No	No	No

VARSHA advantage: Combines the best features of all competitors while adding unique innovations — shield accumulation, RainBoost, TipPoints, and the 5-signal fraud detection system — none of which exist in any competing product.

12. Risk Assessment

12.1 Technical Risks

Risk	Probability	Impact	Mitigation
Weather API downtime	Medium	High	Redundant APIs — IMD primary + OpenWeatherMap backup
Fraud patterns evolving	High	Medium	Weekly ML retraining + continuous monitoring
Payment gateway failure	Low	High	Multiple UPI providers + retry logic
Database outage	Low	High	PostgreSQL replication + daily backups

12.2 Operational Risks

Risk	Probability	Impact	Mitigation
Partner adoption low	Medium	High	Pilot programme + referral incentives + Swiggy/Zomato partnership
Regulatory compliance	Low	Medium	Built on existing IRDAI parametric insurance framework
Claim disputes	Medium	Medium	Transparent trigger data available on partner dashboard
Premium collection fail	Low	Medium	Auto-deduction model + weekly cap limits loss exposure

12.3 Market Risks

Risk	Probability	Impact	Mitigation
Competitors copy model	High	Medium	Patent pending + first-mover advantage + Chennai-specific depth
Climate pattern changes	Medium	Medium	Continuous trigger recalibration with rolling weather data
Economic downturn	Medium	Low	Pay-as-you-earn model naturally reduces cost during low-income periods

13. Regulatory Compliance

13.1 IRDAI Framework Alignment

VARSHA operates under the parametric insurance framework recognised by IRDAI:

IRDAI requirement	VARSHA compliance
Clear trigger definition	20mm / 6hr rain · 40 °C heat · 1.5m flood level · official curfew notice
Transparent payout	₹2,000 fixed payout · 90-second SLA · no discretionary adjustment
Data source reliability	IMD (government) + Chennai Corporation sensors (public infrastructure)
No health / accident coverage	Strictly income loss only — no vehicle repair, health, or accident payouts
Weekly pricing structure	₹75 maximum weekly cap — fully aligned with IRDAI weekly basis requirement

13.2 Data Privacy

- Partner data stored with end-to-end encryption
- Location data used exclusively for claim validation — never for profiling
- No partner data sold to third parties
- GDPR-style informed consent collected at onboarding for all data uses

13.3 Future Government Integration Roadmap

Programme	Integration opportunity
Tamil Nadu Gig Worker Welfare Board	Direct policy alignment; government co-branding potential
ESIC (Employees' State Insurance)	Complementary coverage — VARSHA covers income, ESIC covers health
PMSYM (PM Shram Yogi Maandhan)	Shared worker identity layer; premium subsidy possibility

15. Conclusion

VARSHA represents a fundamental rethink of how insurance works for gig workers. By aligning payment with earnings, providing free coverage after a threshold, and delivering instant payouts with zero paperwork, VARSHA addresses the three biggest barriers that keep delivery partners uninsured: affordability, accessibility, and trust.

The 5-signal fraud detection system ensures that while genuine partners like Karthik receive their money in 90 seconds, bad actors cannot exploit the system. The RainBoost feature helps partners recover lost income faster, and TipPoints engages customers as part of the solution rather than bystanders.

VARSHA is not just insurance. It's financial dignity for those who deliver our food every day.